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GOSFORD HIGH SCHOOL

2015

Higher School Certificate

MATHEMATICS

Assessment Task 2

General Instructions

- Reading Time 5 minutes
- Working Time 90 minutes
- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is provided
- Use multiple choice answer sheet provided for Section 1 (Q 1 to 5)
- For Section 2 (Q 6 – 9) show relevant mathematical reasoning and/or calculations
- Total marks 65

Use the multiple choice answer sheet provided

1. Which of the following is NOT equal to $\tan \phi$

(A) $\sec^2 \phi - 1$

(B) $\cot(90^\circ - \phi)$

(C) $\tan(180^\circ + \phi)$

(D) $\frac{\sin \phi}{\cos \phi}$

2. The range of the function $y = \sqrt{4 - x^2}$ is

(A) $y \geq 0$

(B) $-2 \leq y \leq 2$

(C) $y \leq 2$

(D) $0 \leq y \leq 2$

3. The equation $(2\cos x + 1)\sin x = 0$ for $0^\circ \leq x \leq 360^\circ$ has

(A) 2 solutions

(B) 3 solutions

(C) 4 solutions

(D) 5 solutions

4. The equation of a parabola with focus $(0, -1)$ and focal length 2 units has four possibilities. Which of the following could be the equation of this parabola.

(A) $x^2 = 8(y + 1)$

(B) $(y + 1)^2 = 8(x + 2)$

(C) $x^2 = -8(y + 1)$

(D) $y^2 = 8(x - 3)$

5. Consider the continuous function $y = f(x)$ in the interval $a \leq x \leq b$. It is known that $f(x)$, $f'(x)$ and $f''(x)$ are all positive for all values of x in the given domain.

The **exact area** under the curve and above the x axis between the ordinates at $x = a$ and $x = b$ is defined to be (E) square units.

A student correctly uses the Trapezoidal rule to approximate the given area to be (M) square units.

Which one of the following statements is true

- (A) $E > M$
- (B) $E < M$
- (C) $E = M$
- (D) None of the above, as the number of strips used by the student would need to be known to make the above comparisons between E and M.

Section II

Question 6

(15 marks)

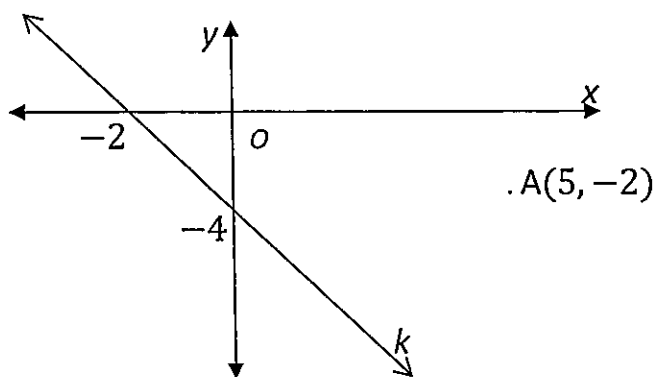
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a) Evaluate $\sqrt{\frac{3.4}{5.6 \times 10^{-9}}}$, writing your answer correct to 3 significant figures. (1)

b) Find integers a and b such that $(3 - \sqrt{2})^2 = a - b\sqrt{2}$ (2)

c) Solve $3x^2 - x - 1 = 0$ writing your answers in exact form. (2)

d) The point A and the line k are shown on the number plane below



(i) Find the equation of the line k , writing your answer in general form (2)

(ii) Find the perpendicular distance from the point $A(5, -2)$ to the line k .
Leave your answer in surd form. (2)

e) $G(x) = 3x^2 - px + p - 3$, where p is a Real constant.

Find p if $G(x)$ has

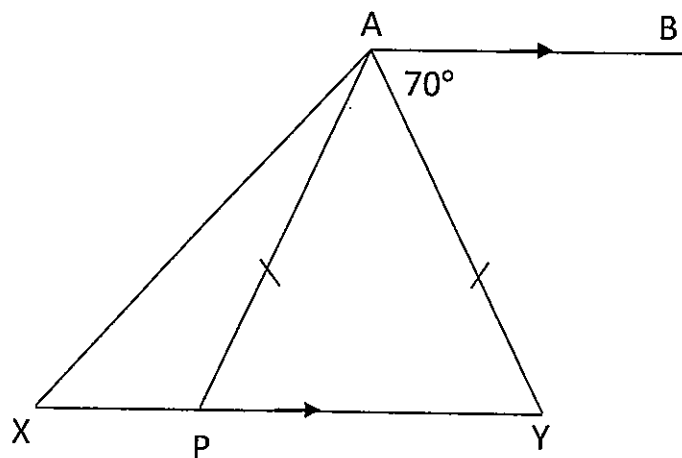
(i) one root equal to 2 (1)

(ii) roots which are reciprocals (2)

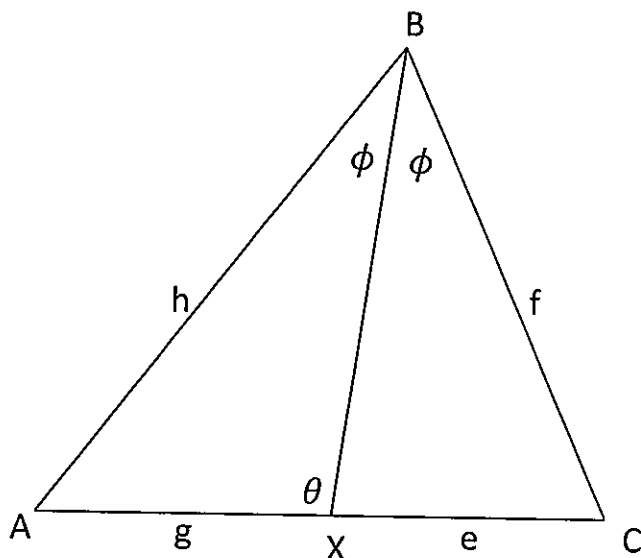
(f) In the diagram $AB \parallel XY$, $\angle BAY = 70^\circ$ and P lies on XY such that $AP = AY$.

Find $\angle PAY$, giving reasons.

(3)



- a) If $\sec\beta = -\frac{7}{5}$ and $\tan\beta > 0$, find the exact value of $\sin\beta$ (2)
- b) Show that $\frac{d}{dx}[x\sqrt{1-2x}] = \frac{1-3x}{\sqrt{1-2x}}$ (3)
- c) Find the equation of the tangent to the parabola $y = x^2 - 2x + 4$ at the point where the tangent is parallel to the line $y = 2x + 1$ (3)
- d) Given that m is a real number, show that the line $y = mx - 3m^2$ touches the parabola $x^2 = 12y$ (3)
- e) In triangle ABC, $\angle BXA = \theta$, $AX = g$, $CX = e$, $AB = h$, $BC = f$ and BX bisects $\angle ABC$ such that $\angle ABX = \angle CBX = \phi$.



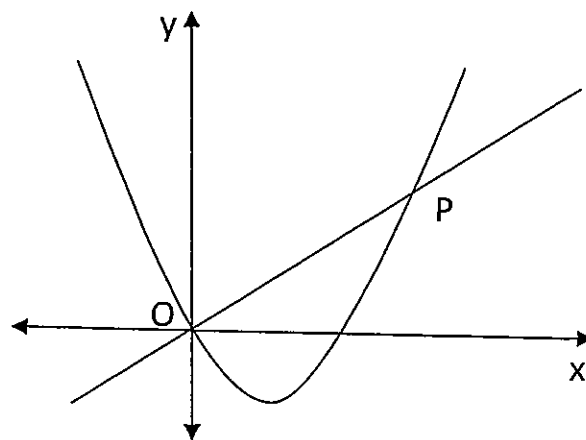
- (i) With consideration to $\triangle ABX$, show that $\frac{\sin\phi}{\sin\theta} = \frac{g}{h}$ (1)
- (ii) Hence, and with similar consideration to $\triangle CBX$, prove that $\frac{e}{f} = \frac{g}{h}$ (3)

(a) Find $\int (1 + 2x)^3 dx$ (1)

(b) Evaluate $\int_1^4 \frac{x+1}{\sqrt{x}} dx$ (3)

- (c) The area bounded by the curve $y = x^2 + 1$, the coordinate axes and the line $x = 4$ is rotated about the x axis.
Find the volume of the solid generated. (4)

- (d) The straight line $y = 3x$ and the parabola $y = 2x^2 - 5x$ meet at the points O and P as indicated on the number plane below.



- (i) Find the coordinates of P (1)

- (ii) Find the area enclosed between the line and the parabola (3)

- a) A curve is defined by the equation $y = 7 + 4x^3 - 3x^4$
- i) Find the coordinates of the two stationary points on the curve. (2)
- ii) Find all values of x for which $\frac{d^2y}{dx^2} = 0$ (2)
- iii) Determine the nature of the stationary points. (3)
- iv) Sketch the curve in the domain $-1 \leq x \leq 2$ (3)

b) **NOTE**

For this question students are given the following two formulae

$$\text{Volume of the Cylinder } V = \pi r^2 h$$

$$\text{Surface Area of the Cylinder } S = 2\pi r^2 + 2\pi r h$$

A closed cylinder, with radius (r) and height (h) has a volume (V) of 2156 cm^3 .

- (i) Show that the surface area(S) of the cylinder is given by $S = 2\pi r^2 + \frac{4312}{r}$. (1)
- (ii) Hence, using differential calculus and *the approximation* $\pi = \frac{22}{7}$,
find the radius (r) so that the surface area of the cylinder is a minimum. (4)

- e) (i) Copy and complete the following table for $y = 2^x - 1$

x	0	1	2	3	4
y					

(1)

- (ii) Use Simpson's Rule with 5 function values to evaluate

$$\int_0^4 (2^x - 1) dx$$

(2)

NAME _____

ANSWER SHEET FOR SECTION 1 - Multiple Choice

1.	A	B	C	D
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2.	A	B	C	D
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3.	A	B	C	D
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4.	A	B	C	D
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5.	A	B	C	D
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Table of Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

e) (i) $y = 2^x - 1$

x	0	1	2	3	4
y	0	1	3	7	15

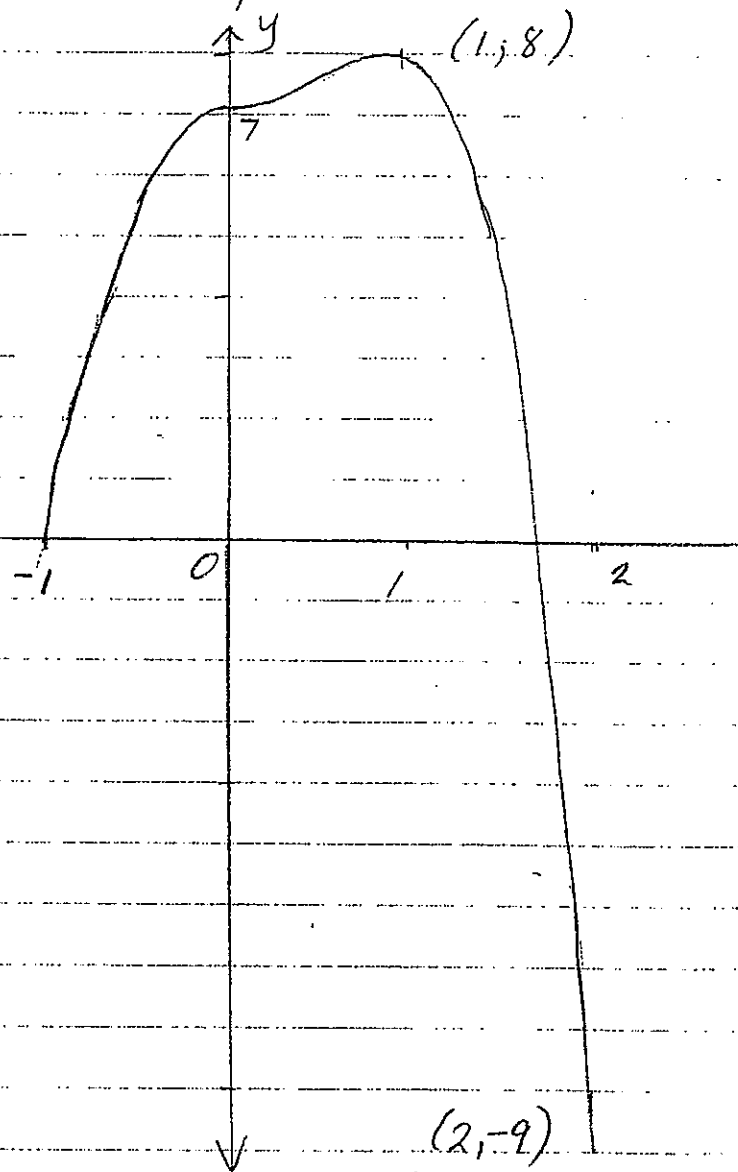
i) $\int_0^4 (2^x - 1) dx = \frac{1}{3} [0 + 15 + 2 \times 3 + 4(1 + 7)]$
 $= \frac{1}{3} [53]$

$= \frac{53}{3}$

$\therefore$ concavity change
 $\therefore (0, 7)$ is a horizontal point of inflexion
 when $x = 1$, $\frac{d^2y}{dx^2} < 0$

$\therefore (1, 8)$ is a maximum turning point

(iv) $f(-1) = 0$
 and $f(2) = -9$



Question 9

a) (i) $y = 7 + 4x^3 - 3x^4$

$\frac{dy}{dx} = 12x^2 - 12x^3$

For stationary points $\frac{dy}{dx} = 0$

$12x^2 - 12x^3 = 0$

$12x^2(1 - x) = 0$

$\therefore x = 0, 1$

Stat. pts at $(0, 7)$ & $(1, 8)$

(ii) $\frac{d^2y}{dx^2} = 24x - 36x^2$

$24x - 36x^2 = 0$

$12x(2 - 3x) = 0$

$x = 0, \frac{2}{3}$

(iii) when $x = 0$, $\frac{d^2y}{dx^2} = 0$

when $x < 0$, say $x = -1$, $\frac{d^2y}{dx^2} < 0$

when $x > 0$, say $x = \frac{1}{2}$, $\frac{d^2y}{dx^2} > 0$

Question 9 (continued)

oops!

$$(i) \quad 2156 = \pi r^2 h$$

$$h = \frac{2156}{\pi r^2}$$

$$\therefore S = 2\pi r^2 + 2\pi r \cdot \frac{2156}{\pi r^2}$$

$$S = 2\pi r^2 + \frac{4312}{r}$$

$$(ii) \quad S = 2\pi r^2 + 4312r^{-1}$$

$$\frac{ds}{dr} = 4\pi r - 4312r^{-2}$$

$$= 4\pi r - \frac{4312}{r^2}$$

$$\frac{d^2s}{dr^2} = 4\pi + 8624r^{-3}$$

$$= 4\pi + \frac{8624}{r^3}$$

$$\therefore \frac{d^2s}{dr^2} > 0 \text{ for all } r > 0$$

$$\text{For stat. pts } \frac{ds}{dr} = 0$$

$$\therefore 4\pi r - \frac{4312}{r^2} = 0$$

$$\therefore 4\pi r^3 - 4312 = 0$$

$$r^3 = \frac{4312}{4\pi}$$

$$r^3 = 343 \text{ if } \pi = \frac{22}{7}$$

$$r = 7$$

$$\therefore \text{Now since } \frac{d^2s}{dr^2} > 0$$

$\therefore$ A minimum value of Surface Area (S) occurs when $r = 7$

$$8c) \quad V = \pi \int_0^4 (x^2 + 1)^2 dx$$

$$V = \pi \int_0^4 (x^4 + 2x^2 + 1) dx$$

$$= \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^4$$

$$= \pi \left[\frac{4^5}{5} + \frac{2}{3} \times 4^3 + 4 - 0 \right]$$

$$= \frac{3772\pi}{15} \text{ cubic units}$$

Solutions

Section I

- (1) A (2) D (3) D
(4) B (5) B

Section II

Question 6

a) 24600 or 2.46×10^4

b) $(3 - \sqrt{2})^2 = 9 + 2 - 6\sqrt{2}$
 $= 11 - 6\sqrt{2}$

$\therefore a = 11, b = 6$

c) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{1 \pm \sqrt{(-1)^2 - 4(3)(-1)}}{6}$

$x = \frac{1 \pm \sqrt{13}}{6}$

d) (i) $M_R = \frac{4}{-2} = -2$

$\therefore$ Equation of line k is

$y = -2x - 4$

or $2x + y + 4 = 0$ in general form

(ii) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$d = \frac{|2(5) + 1(-2) + 4|}{\sqrt{(2)^2 + (1)^2}}$

$d = \left| \frac{12}{\sqrt{5}} \right|$

$d = \frac{12}{\sqrt{5}}$ or $\frac{12\sqrt{5}}{5}$

e) $G(2) = 0$

$\therefore 3(2)^2 - 2p + p - 3 = 0$

$\therefore p = 7$

(ii) Product of Roots = 1

$\therefore \frac{p-3}{3} = 1$

$\therefore p = 6$

f) $\hat{A}YP = \hat{B}AY = 70^\circ$

(alternate angles, $AB \parallel XY$)

$\hat{A}PY = \hat{A}YP = 70^\circ$

(equal angles opposite equal sides of an isosceles triangle)

$\hat{P}AY = 180^\circ - (\hat{A}YP + \hat{A}PY)$

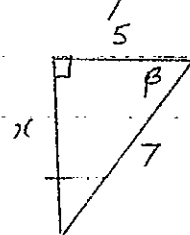
(angle sum of a triangle)

$\therefore \hat{P}AY = 180^\circ - 2 \times 70^\circ$
 $= 40^\circ$

Question 7

a) $\cos \beta = -\frac{5}{7}$ & $\tan \beta > 0$

$\therefore \beta$ lies in 3rd quadrant



$x^2 = 7^2 - 5^2$

$x^2 = 24$

$x = 2\sqrt{6}$

$\therefore \sin \beta = -\frac{2\sqrt{6}}{7}$

b) Let $y = x(1-2x)^{1/2}$

$\frac{dy}{dx} = (1-2x)^{1/2} \cdot 1 + x \cdot \frac{1}{2} (1-2x)^{-1/2} \cdot (-2)$

$= (1-2x)^{1/2} - x(1-2x)^{-1/2}$

$= (1-2x)^{-1/2} [(1-2x) - x]$

$= \frac{1-3x}{\sqrt{1-2x}}$

c) $\frac{dy}{dx} = 2x - 2$

$\therefore 2x - 2 = 2$ at pt of contact

Question 7(c) continued

$$\therefore x = 2$$

$$\text{when } x = 2, y = (2)^2 - 2(2) + 4 = 4$$

$\therefore (2, 4)$ is point of contact

$\therefore$ Equation of line is

$$y - 4 = 2(x - 2)$$

$$y - 4 = 2x - 4$$

$$y = 2x$$

d) Solving simultaneously

$$x^2 = 12(mx - 3m^2)$$

$$x^2 = 12mx - 36m^2$$

$$x^2 - 12mx + 36m^2 = 0$$

$$\Delta = b^2 - 4ac$$

$$= (-12m)^2 - 4(1)(36m^2)$$

$$= 144m^2 - 144m^2$$

$$= 0$$

$\therefore$ One solution

$\therefore y = mx - 3m^2$ is

a tangent to the curve for all real m

$$e) (i) \frac{\sin \phi}{g} = \frac{\sin \theta}{h}$$

$$\frac{\sin \phi}{\sin \theta} = \frac{g}{h}$$

$$(ii) \frac{\sin \phi}{e} = \frac{\sin(180-\theta)}{f}$$

$$\frac{\sin \phi}{e} = \frac{\sin \theta}{f}$$

$$\frac{\sin \phi}{\sin \theta} = \frac{e}{f}$$

$$\therefore \frac{e}{f} = \frac{g}{h} \left(= \frac{\sin \phi}{\sin \theta} \right)$$

Question 8

$$a) \int (1+2x)^3 dx = \frac{(1+2x)^4}{4 \times 2} + C = \frac{(1+2x)^4}{8} + C$$

$$b) \int_1^4 \frac{x+1}{x^{\frac{1}{2}}} dx = \int_1^4 x^{\frac{1}{2}} + x^{-\frac{1}{2}} dx = \left[\frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} \right]_1^4 = \left[\frac{2}{3} \times 8 + 2 \times 2 - \left(\frac{2}{3} + 2 \right) \right] = \frac{20}{3}$$

c) oops $\rightarrow$ see back page

$$d) \text{ For } P, 3x = 2x^2 - 5x \\ 2x^2 - 8x = 0 \\ 2x(x - 4) = 0 \\ x = 0, 4$$

$$P(4, 12)$$

$$(ii) A = \int_0^4 3x - (2x^2 - 5x) dx = \int_0^4 8x - 2x^2 dx = \left[4x^2 - \frac{2x^3}{3} \right]_0^4 = \left[64 - \frac{128}{3} \right] = \frac{64}{3} \text{ sq. units}$$